Lesson 01 Demo 01

Creating and Using a JSON Server File

Objective: To create a JSON file with sample user data, serve it as a REST API using json-server, and retrieve the data using curl to simulate API interactions for frontend development and testing without a backend server

Tools required: Visual Studio Code and json-server

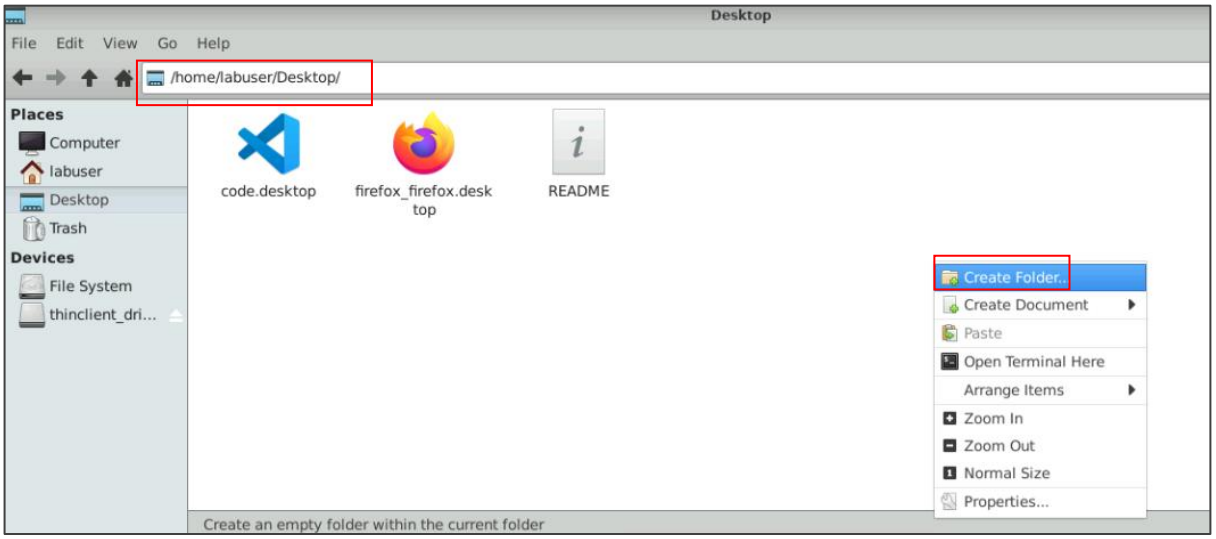
Prerequisites: None

Steps to be followed:

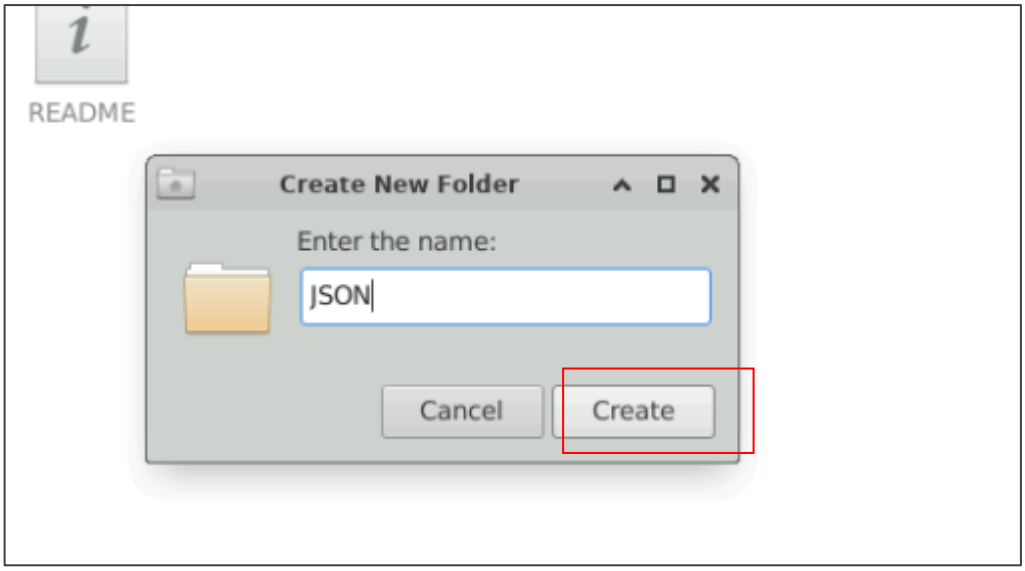
1. Create a db.json file with user data
2. Install and run json-server
3. Use curl to retrieve and display the user data

Step 1: Create a db.json file with user data

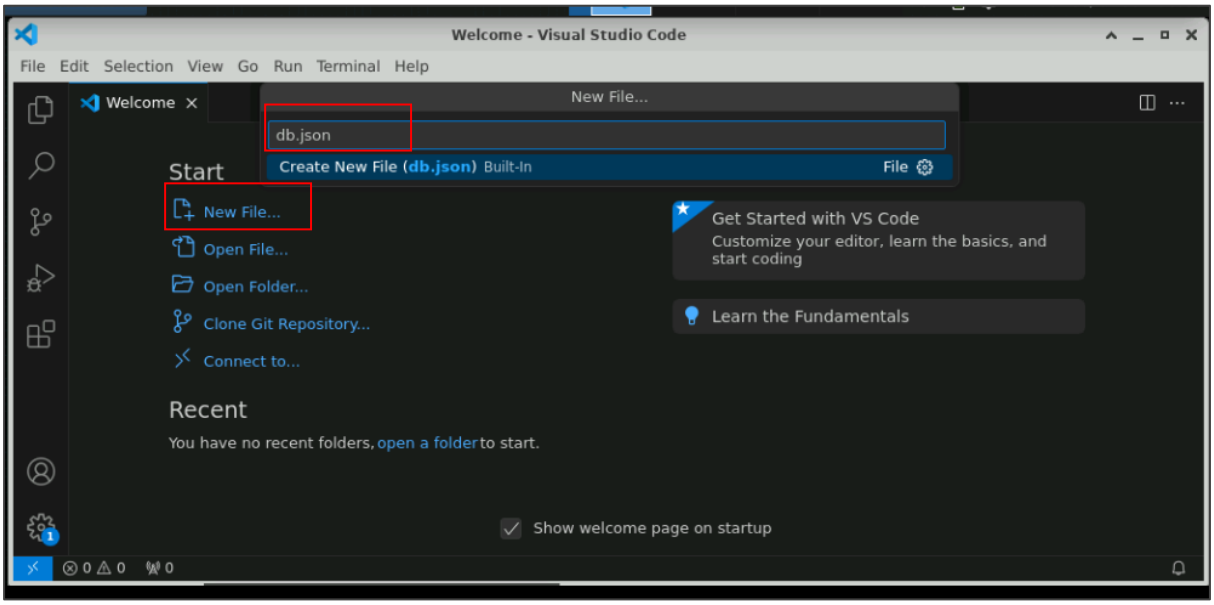
1.1 Right-click on an empty space on your Desktop, then click **Create Folder**

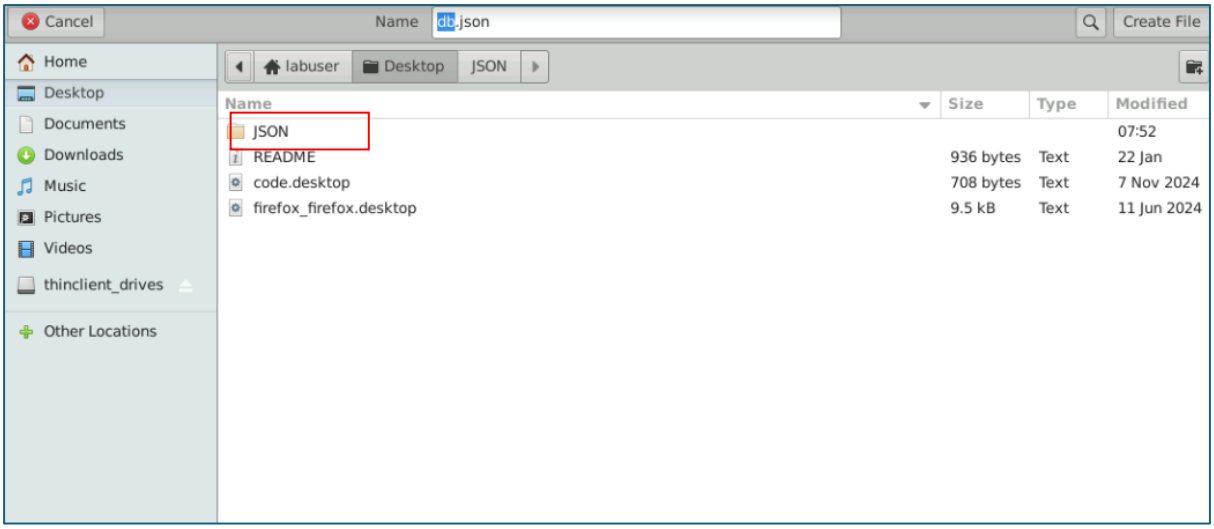


1.2 Name the folder **JSON** and click **Create** to confirm



1.3 Open Visual Studio Code and create a **New File…** named **db.json** inside the **JSON** folder

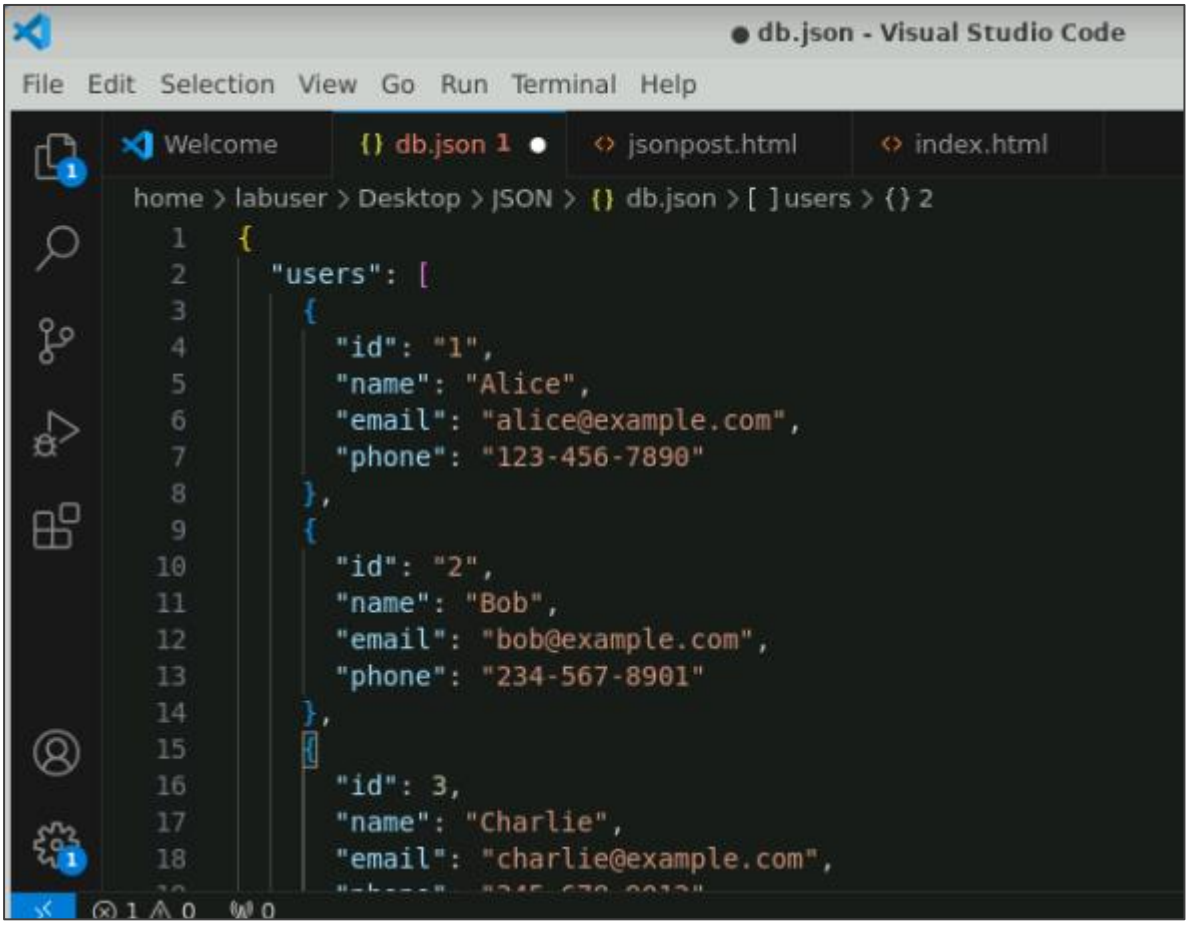




1.4 Copy and paste the following JSON structure into the db.json file:

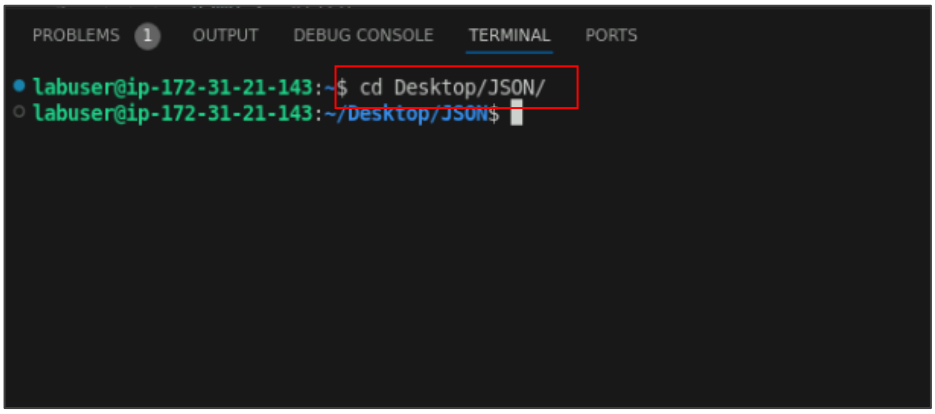
```
{
"users": [
{
  "id": 1,
  "name": "Alice",
  "email": "alice@example.com",
  "phone": "123-456-7890"
 },
 {
  "id": 2,
  "name": "Bob",
  "email": "bob@example.com",
  "phone": "234-567-8901"
 },
 {
  "id": 3,
  "name": "Charlie",
  "email": "charlie@example.com",
  "phone": "345-678-9012"
 }
]
```

}



Step 2: Install and run json-server

2.1 Open the terminal and run the following command to navigate to the JSON folder:
cd Desktop/JSON/



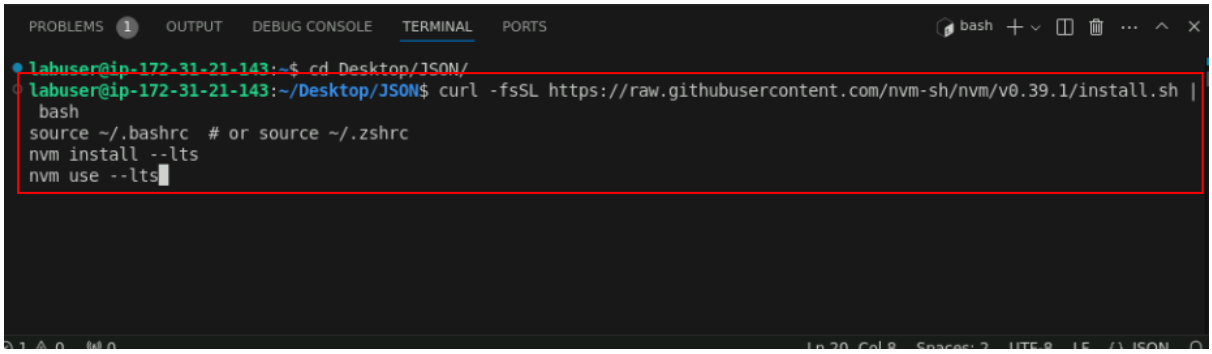
2.2 Run the following commands to install nvm (Node Version Manager) and the latest LTS version of Node.js:

curl -fsSL https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.1/install.sh | bash

source ~/.bashrc # or source ~/.zshrc

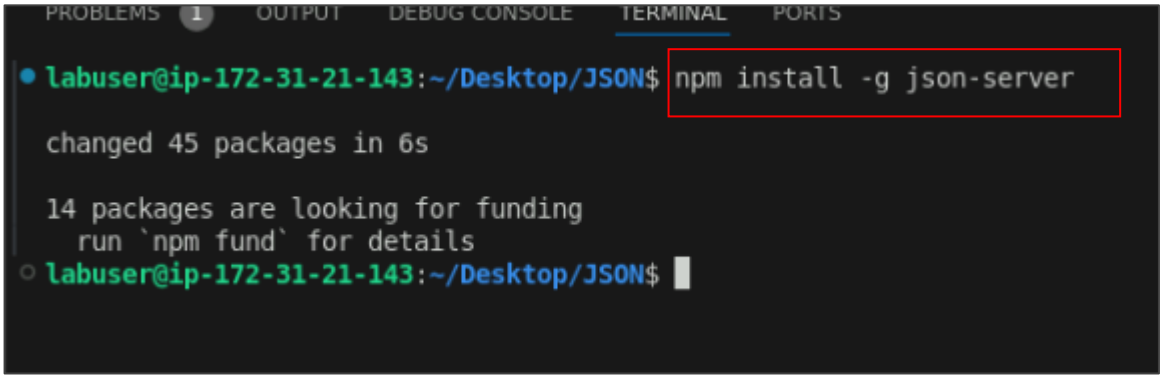
nvm install --lts

nvm use --lts



2.3 Run the following command to install json-server:

npm install -g json-server



2.4 Run the following command to serve the JSON data:

json-server --watch db.json --port 3001

```
labuser@ip-172-31-21-143:~/Desktop/JSON$ json-server --watch db.json --port 3001
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3001
Press CTRL-C to stop
Watching db.json...

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Index:
http://localhost:3001/

Static files:
Serving ./public directory if it exists

Endpoints:
http://localhost:3001/users
```
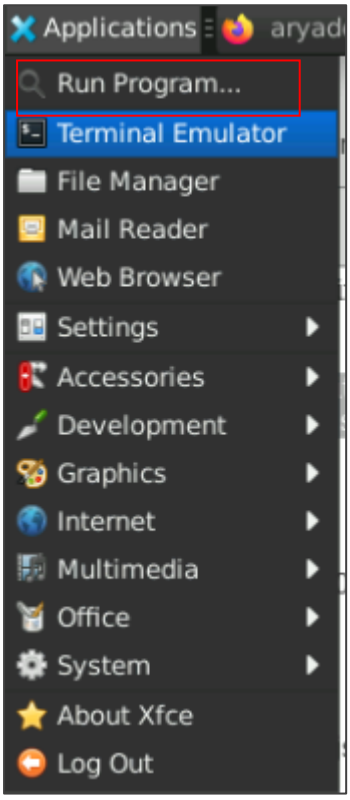

Note: If port 3001 is already in use, you can change it to 3002, 3003, or any other available port.

Step 3: Use curl to retrieve and display the user data

3.1 Open a new terminal window, keep the json-server running, and open another terminal



3.2 Run the following command to fetch and display user data from the server:
curl http://localhost:3001/users

```
labuser@ip-172-31-21-143: ~
File  Edit  View  Search  Terminal  Help
labuser@ip-172-31-21-143:~$ curl http://localhost:3001/users
[
  {
    "id": "1",
    "name": "Alice",
    "email": "alice@example.com",
    "phone": "123-456-7890"
  },
  {
    "id": "2",
    "name": "Bob",
    "email": "bob@example.com",
    "phone": "234-567-8901"
  },
  {
    "id": "3",
    "name": "Charlie",
    "email": "charlie@example.com",
    "phone": "345-678-9012"
  }
]labuser@ip-172-31-21-143:~$
```

By following these steps, you have successfully created a JSON file with sample user data, served the data as a REST API using json-server, and retrieved it using curl. This exercise emphasizes the value of simulating APIs for frontend development and testing, enabling efficient API interaction without the need for a backend server.